Aarti Kumari Singh

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SUMMARY

Final-year Information Technology engineering student (CGPA 9.33) who builds backend systems and data-driven applications with Python, Java, SQL and Node.js. Built a churn-prediction model with a Flask dashboard and a Firebase-based food donation platform, and contributed to an AI-powered ticketing app as a project mentee at Persistent Systems. Seeking internships in software development, backend development or data analytics.

EXPERIENCE

Project Mentee, SupportSenseAI Aug 2026 – Oct 2026
Persistent Systems | 8-week mentorship program

- Contributed, as part of a team, to the backend of an AI-powered customer support ticket management application built with **Node.js, Express and PostgreSQL**, covering REST APIs, routes, controllers, models and ticket workflows.
- Worked on password hashing with **bcrypt** for secure credential storage and tested the API using **Thunder Client**.
- Tested ticket workflows, **JWT authentication** and **role-based access control (RBAC)** permissions across **3 user roles** (Customer, Agent, Admin).
- Performed QA and security checks on internal Agent note visibility, input handling, search/filter functionality and **rate limiting**.
- Ran team-written **Jest tests** for authentication, RBAC and controller behavior; used **Docker and Docker Compose** to run and test the application locally.
- Learned how the Node.js backend integrates with a **Python FastAPI** AI microservice powering **4 AI-assisted workflows**: ticket triage, department assignment, response verification and timeline summarization.

TECHNICAL PROJECTS

Customer Segmentation & Retention Analysis System Python, Pandas, Matplotlib, Flask, HTML

- Built a **Random Forest model** achieving **96.27% accuracy** for telecom churn prediction, analyzing **667+ customer records** to uncover behavioral patterns.
- Identified **92 high-risk customers** and a **14% churn rate** to support targeted retention strategies and data-driven decisions.
- Applied **KMeans clustering** to segment customers into **3 groups** (low, medium, high value) and developed a **Flask-based dashboard** to visualize churn predictions and customer insights.

Food Waste Management System (MealBridge) HTML, CSS, JavaScript, Firebase

- Developed a food donation web platform connecting donors and NGOs, using **Firebase** and JavaScript for real-time data storage and retrieval (tested with 10+ records).
- Implemented a role-based interface (Donor/NGO) and **CRUD operations**, reducing manual coordination effort by ~40% through streamlined food pickup tracking.

SKILLS

Languages & Core: Python, Java, C++, C, PHP, SQL, JavaScript; DSA, OOP, DBMS
Web & Databases: HTML, CSS, Flask, MySQL, PostgreSQL, Firebase
Data & ML: Pandas, Matplotlib, Random Forest, KMeans clustering
Backend & QA: Node.js, Express, REST APIs, JWT, bcrypt, RBAC, Jest, Docker, Docker Compose, FastAPI (exposure)
Tools: Git, GitHub, VS Code, Figma, Cursor AI, Thunder Client

EDUCATION

Bachelor of Engineering, Information Technology (Final Year) Present
Shree Rayeshwar Institute of Engineering and Information Technology, Goa | CGPA: 9.33

H.S.S.C, Madkaikar Higher Secondary School of Arts and Science, 80% 2023

S.S.C, Madkaikar Navchaitanya High School, 87.7% 2021

ACHIEVEMENTS, CERTIFICATIONS & WORKSHOPS

- Completed **Introduction to Generative AI Studio** (Google Cloud course via Simplilearn SkillUP), August 2026.
- Participated in **Smart India Hackathon (SIH) 2026**. Previously presented “AI-Driven Virtual Interview Platform for Automated Candidate Assessment and Feedback” at SIH as part of Team Knowledge Crew.
- Competed in **2 coding competitions** and designed UI/UX prototypes in **4 Figma design competitions**.
- Completed workshops on ReactJS, Cursor AI and Umbraco; contributed to organizing college events.

LANGUAGES

English, Hindi, Konkani, Marathi